

Deepak Mallampati

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US-based, F-1 OPT, open to Greater Boston relocation + sponsorship

PROFESSIONAL SUMMARY

Backend-first Applied AI engineer (2.5+ yrs) with production RAG, LLM agents, and full-stack delivery. Stack 1:1 with Newton Research's: Python + LangChain + LangGraph + sentence-transformers + RAGAS-style eval discipline + React 19/TypeScript/Vite. Production proof: ~35% retrieval precision gain on healthcare RAG, 5-stage agentic claims pipeline with schema-validated JSON contracts and audit traces, multi-provider LLM integration shipped to Chrome Web Store, and ~30% post-deploy defect reduction. Master's degree (Kent State University). Comfortable in HIPAA-conscious environments and ships at the eval-and-grounding discipline level.

CORE COMPETENCIES

Python Backend Engineering • LLM APIs (OpenAI / Anthropic / Google) • LangChain + LangGraph (Agent Orchestration) • RAG + Sentence-Transformers + Semantic Retrieval • Eval Discipline (RAGAS-Style Grounded Metrics) • Schema-Validated Multi-Agent Pipelines • Postgres / SQL / Pandas / NumPy • React 19 + TypeScript + Vite • Docker + CI/CD + Structured Logging • scikit-learn / XGBoost / Time-Series Forecasting

PROFESSIONAL EXPERIENCE

Progress Solutions Inc.

Jul 2025 - Present

Jr. AI/ML Engineer Trainee - USA - Healthcare AI / Applied AI Consulting

- Built **RAG systems** for clinical knowledge retrieval using recursive semantic chunking and transformer embeddings; ~**35%** retrieval precision gain, >**30%** irrelevant retrieval reduction, retrieval-grounded alignment >**90%** in eval.
- Developed **agentic LLM workflows** with structured reasoning, tool discipline, and grounding rules (LangChain-class patterns); ~**25%** agent response stability gain.
- Shipped **predictive ML pipelines** (scikit-learn, XGBoost) for patient no-show, care engagement, and prioritization; **15-20%** recall gains on high-risk categories with precision >**90%**.
- Packaged ML/LLM inference as **FastAPI/Flask** services in **Docker** with structured logging and load simulation; ~**30%** post-deploy defect reduction.
- Built preprocessing pipelines (Pandas, NumPy) for EHR extracts, appointment histories, support ticket logs; raised dataset reliability above **98%** and cut downstream model instability ~**40%**.
- Maintained HIPAA-conscious data governance: de-identification, lineage docs, evaluation audit trails.

Energy Solutions International (Emerson)

Jun 2022 - Apr 2023

Database & DevOps Performance Engineer (Intern) - Hyderabad, India - Oil & Gas / Enterprise ERP

- Built **CI/CD pipelines with Jenkins** for schema/SP deployments with rollback checkpoints; >**30%** deployment error reduction, ~**35-40%** release cycle improvement.
- Tuned T-SQL stored procedures for Synthesis Order-to-Cash (contracts/nominations/allocations/invoicing); ~**20%** query time reduction, ~**25%** latency reduction.
- Built SQL Server DMV + Grafana dashboards; ~**25%** incident recurrence cut, ~**30%** faster RCA.
- Supported RBAC + audit logging for financial modules in compliance-sensitive environment.

Suvidha Foundation

Jan 2022 - Mar 2022

Machine Learning Engineer - Hyderabad, India - Nonprofit Tech / Video Analytics / Applied NLP

- Shipped **transformer-based video summarization** across 5,000+ sessions; **60-70%** review time reduction, ~**85%** alignment with human curation.
- Implemented **document Q&A with RAG** (semantic chunking + embedding retrieval); ~**30%** hallucination reduction, >**85%** contextual accuracy.
- Deployed as **Flask API** with lightweight web UI for non-technical staff.

PROJECTS

Agentic Healthcare Claims Processing & Fraud Risk Intelligence System

Multi-agent pipeline (Intake → Validation → Consistency → Duplicate → Risk Score) with schema-validated JSON contracts to prevent cascading hallucinations. RAG-grounded CPT/ICD validation against vector-indexed policy documents; ANN similarity for duplicates; explainable risk scoring with audit-ready reasoning traces.

Manga Lens - Chrome Extension for Real-Time AI Manga/Webtoon Translation Shipped

Solo-shipped Manifest V3 extension integrating 4 AI vision providers (Claude Sonnet, GPT-4o mini, GPT-4.1 Nano, Ollama/minicpm-v). Per-provider payload normalization (WebP cloud / JPEG Ollama), 7-day translation cache, per-domain configs for 29 sites, narrowed host permissions.

Dream Decoder - AI-Powered Multimodal Creative Intelligence Platform

Full-stack FastAPI + React/TypeScript/Vite/Tailwind interpreting dreams and synthesizing dreamscapes via Perplexity Sonar + Replicate. Structured prompt transformation layers improved contextual alignment ~30% and first-pass image success ~25-30% over naive direct-prompt orchestration.

IoT + ML Smart Building Indoor Temperature Prediction System

Arduino + DHT11 + DS3231 sensor pipeline with PLX-DAQ logging; time-series forecasting with walk-forward validation comparing Random Forest, SVR, and Linear Regression. Lag-window feature engineering (n_in=6); Random Forest achieved ~20-30% lower MAE vs linear baseline.

Driver Drowsiness Detection System - Real-Time YOLOv8 Fatigue Monitoring

Replaced two-stage CNN pipeline with unified YOLOv8; ~30% inference latency reduction. Sliding-window confidence aggregation suppressed blink false positives ~25%; adaptive frame skipping and NMS tuning for stable real-time operation.

Geospatial Road Network Analysis & Community Detection System

Parsed OpenStreetMap XML into a NetworkX DiGraph; Haversine edge weighting; Dijkstra shortest-path routing; betweenness centrality on 20k-node subgraph; Louvain community detection and MST extraction. ~30% runtime reduction via subgraph scoping.

EDUCATION

Master's Degree - Kent State University, Ohio, USA

Aug 2023 - May 2025

CERTIFICATIONS

TECHNICAL SKILLS

Languages & Backend: Python, FastAPI, Flask (Django-adjacent), SQL (PostgreSQL, T-SQL, SQLite), TypeScript, React, C++ (Arduino), PHP

LLM & RAG: LangChain, LangGraph, sentence-transformers, RAGAS-style grounded eval, OpenAI/Anthropic/Google APIs, RAG, structured outputs, prompt engineering, vector search, embeddings

ML & CV: scikit-learn, XGBoost, PyTorch, Hugging Face Transformers, Diffusers, YOLOv8, OpenCV, ControlNet

Data & Analytics: Pandas, NumPy, feature engineering, time-series forecasting, walk-forward validation, NetworkX

Infra & DevOps: Docker, Jenkins, GitHub Actions (transferable), Grafana, CI/CD, MLOps/LLMOps, RESTful APIs, observability, structured logging

Domains: Applied AI, Healthcare AI (HIPAA-conscious), RAG / Agentic Workflows, Backend Services, Computer Vision, Multimodal Systems